Optimal control

Model answers

Question 1

a) The Lagrangian is

$$L = \frac{\sqrt{1 + (y'(x))^2}}{y(x)}$$

[3 marks]

b) Since the Lagrangian does not depend on x, we can apply directly the "no x" result, and therefore the Euler-Lagrange equation reduces to

$$L_{y'} y'(x) - L = c_1 \tag{1}$$

where c_1 is a constant. Computing

$$L_{y'} = \frac{y'(x)}{y(x)\sqrt{1 + (y'(x))^2}}$$

and substituting this and the Lagrangian in (1) gives

$$\frac{y'(x)^2}{y(x)\sqrt{1 + (y'(x))^2}} - \frac{\sqrt{1 + (y'(x))^2}}{y(x)} = c_1,$$

which yields

$$\frac{1}{y(x)\sqrt{1 + (y'(x))^2}} = -c_1$$

and can further be rearranged as

$$y'(x) = \pm \frac{\sqrt{c_2 - y(x)^2}}{y(x)} \tag{2}$$

where c_2 is a constant ($= \frac{1}{c_1^2}$).

[6 marks]

c) Equation (2) can easily be integrated as

$$\pm \int \frac{y}{\sqrt{c_2 - y^2}}\, dy = \int dx$$

to give

$$\mp \sqrt{c_2 - y(x)^2} = x + c_3$$

where c_3 is a constant of integration, and finally the general form of the extremal curve is

$$y(x)^2 = -x^2 + c_4 x + c_5 \tag{3}$$

where c_4 and c_5 are constants.

[4 marks]

Answers

d) By using the boundary condition $y(0) = 0$ in (3), gives $c_5 = 0$. Also, since the terminal point varies along the line $y = x - 5 = \Theta(x)$, then

$$y(x_f) = x_f - 5 = \Theta(x_f) \tag{4}$$

with $\frac{d\Theta}{dx}(x_f) = 1$. Therefore, the transversality condition

$$L_{y'}(x_f, y(x_f), y'(x_f)) \left(\frac{d\Theta}{dx}(x_f) - y'(x_f) \right) + L(x_f, y(x_f), y'(x_f)) = 0$$

gives

$$\left(\frac{y'(x_f)}{y(x_f)\sqrt{1 + (y'(x_f))^2}} \right) \left(1 - y'(x_f)\right) + \frac{\sqrt{1 + (y'(x_f))^2}}{y(x_f)} = 0,$$

which simplifies to

$$y'(x_f)(1 - y'(x_f)) + 1 + (y'(x_f))^2 = y'(x_f) + 1 = 0,$$

and finally gives

$$y'(x_f) = -1 \tag{5}$$

By differentiating (3), yields

$$2y(x)y'(x) = -2x + c_4$$

such that

$$2y(x_f)y'(x_f) = -2x_f + c_4$$

and by substituting from (4) and (5) gives $c_4 = 10$. Therefore, the extremal curve is

$$y(x)^2 = -x^2 + 10x$$

[7 marks]

Answers

Question 2

a) The Hamiltonian is

$$H(t, x(t), u(t), p(t)) = p(t)u(t) - \Big((x(t) - \cos t)^2 + u(t)^2\Big)$$

in which it is assumed that $p(t)$ has been normalised so that $p_0 = -1$.

[3 marks]

b) The necessary conditions for optimality are

$$\dot{p}^*(t) = -H_x|_* = 2x^*(t) - 2\cos t \tag{6}$$

and furthermore, since the control values are unconstrained

$$H_u = p^*(t) - 2u^*(t) = 0$$

Note that

$$H_{uu} = -2 < 0$$

therefore,

$$u^*(t) = \frac{p^*(t)}{2} \tag{7}$$

maximises the Hamiltonian.

[5 marks]

Answers

c) The first boundary condition is $x^*(0) = 0$. Since the final state is free, the second boundary condition that must be satisfied is $p^*(t_f) = -\phi_x(x^*(t_f))$ where $\phi = x(t)^2$, and therefore

$$p^*(t_f) = -2x^*(t_f) \tag{8}$$

[4 marks]

d) By differentiating (6) and combining it with the state equation $\dot{x}(t) = u(t)$ and (7), one obtains

$$\ddot{p}^*(t) = 2\dot{x}^*(t) + 2\sin t = 2u^*(t) + \sin t = p^*(t) + 2\sin t$$

which is the differential equation

$$\ddot{p}^*(t) - p^*(t) = 2\sin t$$

and can easily be solved to give

$$p^*(t) = Ae^t + Be^{-t} - \sin t \tag{9}$$

[4 marks]

e) By considering the state equation, and (7) and (9)

$$\dot{x}^*(t) = u^*(t) = \frac{p^*(t)}{2} = \frac{A}{2}e^t + \frac{B}{2}e^{-t} - \frac{\sin t}{2}$$

and therefore

$$x^*(t) = \frac{A}{2}e^t - \frac{B}{2}e^{-t} + \frac{\cos t}{2} \tag{10}$$

By using the first boundary condition $x^*(0) = 0$, gives

$$B = A + 1 \tag{11}$$

By using the second boundary condition given in (8) together with (11), yields

$$Ae^{t_f} + (A+1)e^{-t_f} - \sin t_f = -2\left(\frac{A}{2}e^{t_f} - \frac{A+1}{2}e^{-t_f} + \frac{\cos t_f}{2}\right)$$

and therefore

$$A = \frac{1}{2}\left(\sin t_f - \cos t_f\right)e^{-t_f} \tag{12}$$

$$B = \frac{1}{2}\left(\sin t_f - \cos t_f\right)e^{-t_f} + 1 \tag{13}$$

Finally, from (10), (12) and (13) the optimal state trajectory is given by

$$x^*(t) = \frac{1}{4}(\sin t_f - \cos t_f)e^{t-t_f} - \frac{1}{4}(\sin t_f - \cos t_f)e^{-t-t_f} - \frac{1}{2}e^{-t} + \frac{1}{2}\cos t \tag{14}$$

Also, from the state equation $u^*(t) = \dot{x}^*(t)$, thus

$$u^*(t) = \frac{1}{4}(\sin t_f - \cos t_f)e^{t-t_f} + \frac{1}{4}(\sin t_f - \cos t_f)e^{-t-t_f} + \frac{1}{2}e^{-t} - \frac{1}{2}\sin t$$

[4 marks]

Answers

Question 3

a) The Hamiltonian is

$$H(x(t), u(t), p(t)) = p_1(t)(x_2(t) + u(t)) - p_2(t)u(t) - \frac{1}{2}x_1(t)^2$$

$$= p_1(t)x_2(t) - \frac{1}{2}x_1(t)^2 + (p_1(t) - p_2(t))u(t)$$

in which it is assumed the costates have been normalised so that $p_0 = -1$.

[2 marks]

b) The switching function $\varphi(t)$ is

$$\varphi(t) = p_1^*(t) - p_2^*(t)$$

associated with a specific optimal trajectory.

[2 marks]

c) By considering the maximisation of the Hamiltonian, the optimal control in terms of the switching function is given by

$$u^*(t) = \text{sgn}(\varphi(t)) = \begin{cases} M & \text{if } \varphi(t) > 0 \\ -M & \text{if } \varphi(t) < 0 \\ ? & \text{if } \varphi(t) = 0 \end{cases}$$

Answers

[2 marks]

d) The adjoint equations are

$$\begin{aligned} \dot{p}_1^*(t) &= -H_{x_1}|_* = x_1^*(t) \\ \dot{p}_2^*(t) &= -H_{x_2}|_* = -p_1^*(t) \end{aligned} \tag{15}$$

[2 marks]

e) For a singular optimal control to exist, $\varphi(t)$ must identically vanish ($\varphi(t) \equiv 0$) on some time interval $t \in [t_1, t_2]$, that is if there is a time interval $[t_1, t_2]$ during which $\varphi(t) = 0$ for all $t \in [t_1, t_2]$ then $[t_1, t_2]$ is a singular interval. We investigate this possibility by noting that if $\varphi(t) = 0$ for $t \in [t_1, t_2]$ then

$$\varphi(t) = \dot{\varphi}(t) = \ddot{\varphi}(t) = \ldots = 0, \qquad t \in [t_1, t_2]$$

During a singular interval

$$\varphi(t) = p_1^*(t) - p_2^*(t) = 0 \tag{16}$$

and by differentiating (16)

$$\dot{\varphi}(t) = \dot{p}_1^*(t) - \dot{p}_2^*(t) = 0 \tag{17}$$

where, by using (15), (17) can be rewritten as

$$\dot{\varphi}(t) = x_1^*(t) + p_1^*(t) = 0 \tag{18}$$

Then, by a further differentiation

$$\ddot{\varphi}(t) = \dot{x}_1^*(t) + \dot{p}_1^*(t) = 0$$

and by substitution from (15) and the first state equation, it follows that

$$\ddot{\varphi}(t) = x_2^*(t) + u^*(t) + x_1^*(t) = 0$$

It can be seen that for the control

$$u^*(t) = -(x_1^*(t) + x_2^*(t)) \tag{19}$$

$\ddot{\varphi}(t) = 0$ and all higher-order derivatives of the switching function are obviously 0, hence, $\varphi(t) \equiv 0$ and therefore singular arcs exist. The singular control law is given by (19).

[7 marks]

f) Since the Hamiltonian is not an explicit function of t and t_f is fixed, the Hamiltonian must be constant for an optimal solution:

$$H(x^*(t), u^*(t), p^*(t)) = p_1^*(t)x_2^*(t) - \frac{1}{2}x_1^*(t)^2 + (p_1^*(t) - p_2^*(t))u^*(t) = \text{const.}$$

and on a singular arc, using (16) and (18), gives

$$-x_1^*(t)x_2^*(t) - \frac{1}{2}x_1^*(t)^2 = \text{const.}$$

This represents a one-parameter family of singular arcs (hyperbolas) in the state space. The singular arcs are truncated at

$$|x_1(t) + x_2(t)| = M$$

because $|u(t)| \leq M$.

Answers

[5 marks]

Question 4

a) The Hamilton-Jacobi-Bellman (HJB) differential equation for this problem is

$$-V_t(t,x) = \inf_{u\in U}\left\{x_1(t)^2 + x_2(t)^2 + u(t)^2 + V_{x_1}(t,x)x_2(t) + V_{x_2}(t,x)u(t)\right\}$$

[3 marks]

b) Since the expression inside the infimum is polynomial in u, we can easily find the control that achieves the infimum. By differentiating with respect to u, gives

$$2u(t) + V_{x_2}(t,x) = 0$$

which yields

$$u(t) = -\frac{V_{x_2}(t,x)}{2} \tag{20}$$

Note that by differentiating the expression inside the infimum once again with respect to u, gives $2 > 0$ and, therefore, the control in (20) does achieve the infimum. Substituting the control (20) into the HJB equation, one obtains

$$-V_t(t,x) = x_1(t)^2 + x_2(t)^2 + V_{x_1}(t,x)x_2(t) - \frac{(V_{x_2}(t,x))^2}{4}$$

[4 marks]

Answers

c) There is no terminal cost, the terminal state is fixed and the terminal time is free. Therefore, the boundary condition that must be satisfied by the value function is

$$V(t,x_f) = 0 \qquad \forall t \in [0,\infty)$$

[3 marks]

d) i) The assumed solution $V(t,x) = \sqrt{3}x_1(t)^2 + ax_1(t)x_2(t) + bx_2(t)^2$ satisfies the boundary condition found in part c) for any value of a and b, since

$$V(t,x_f) = \sqrt{3}x_1(t_f)^2 + ax_1(t_f)x_2(t_f) + bx_2(t_f)^2 = 0$$

because $x_1(t_f) = x_2(t_f) = 0$. By differentiating the assumed solution, one obtains

$$V_{x_1}(t,x) = 2\sqrt{3}x_1(t) + ax_2(t)$$

$$V_{x_2}(t,x) = ax_1(t) + 2bx_2(t) \tag{21}$$

and

$$V_t(t,x) = 0$$

Then by substituting these terms in the HJB equation, yields

$$x_1(t)^2 + x_2(t)^2 + \left(2\sqrt{3}x_1(t) + ax_2(t)\right)x_2(t) - \frac{1}{4}\left(ax_1(t) + 2bx_2(t)\right)^2 = 0$$

which gives

$$x_1(t)^2 + x_2(t)^2 + 2\sqrt{3}x_1(t)x_2(t) + ax_2(t)^2 - \frac{a^2}{4}x_1(t)^2 - abx_1(t)x_2(t) - b^2x_2(t)^2 = 0$$

Since this equation must be satisfied for all $x_1(t)$ and $x_2(t)$, its coefficients need to be zero, and therefore

$$1 - \frac{a^2}{4} = 0$$
$$1 + a - b^2 = 0$$
$$2\sqrt{3} - ab = 0$$

which are satisfied by $a = 2$ and $b = \sqrt{3}$. [7 marks]

ii) From (20) and (21) the optimal control law is

$$u^*(t) = -x_1^*(t) - \sqrt{3}x_2^*(t)$$

[3 marks]

Answers